



IDBI INNOVATE 2026

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Knowledge Partner



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Technology Partner



Team Details

- a. **Team name:** ASM FinTech Developer
- b. **Team leader name:** Abhinav Mukwane
- c. **Problem Statement:** Financial Health Score

Brief about the idea

India has 6.4 crore MSMEs contributing ~30% of GDP, yet 80%+ have no formal credit. New-to-Credit (NTC) and New-to-Bank (NTB) enterprises are rejected for missing paperwork — not for being bad borrowers.

FinRiskLensAI — AI-powered MSME Financial Health Score platform (Top-5 Finalist build)

- One consent, five sources: Udyam, GST (7 return types), ITR, bank statements via Account Aggregator, and MCA company & director records
- One score, six pillars: Financial Health Score 0–1000 from a rules engine + ML.NET LightGBM — every score comes with plain-language reasons
- Financial Health Card: score gauge, 6-pillar radar, strengths & risks, lending eligibility (working capital + term loan) and the right product (CGTMSE / MUDRA / standard)
- **NEW — Score History:** every analysis is saved, so the bank sees the trend, not just today's number
- **NEW — Bank Portal (Customer 360):** one screen per MSME for the credit officer — score, GST, bank, ITR, MCA and directors
- Integrity check: GST, ITR and bank credits are cross-checked; mismatches are flagged for officer review, not silently scored
- Built for Bharat: 7 Indian languages (Bhashini), AI assistant, printable & emailable Financial Report; score in seconds; API-ready for LOS / ULI / OCEN / ONDC

Opportunities

How is it different from existing solutions?

- Bureau scores (CIBIL / Experian) need credit history — we score NTC / NTB MSMEs with none, from their own digital footprint
- Five data sources in one consent flow — Udyam, GST, ITR, bank (AA) and MCA — instead of a single-source view
- Score + score history + bank portal: a working system a credit officer can use today, not just a number
- Cross-source check (GST vs ITR vs bank credits) as a built-in integrity signal — no existing tool does this

How does it solve the problem?

- Ends the 'no documents, no credit' cycle — the MSME's existing filings and statements are the documents
- 2–3 weeks of manual appraisal becomes an automated assessment in seconds
- Every score carries dimension-level reasons — the officer sees why, not just what
- Neutral baseline for first-time borrowers — missing history is not treated as bad history

USP of the proposed solution

- Six-pillar score (Revenue, Cash Flow, Transactions, Compliance, Stability, Debt) — a far richer picture than any single metric
- Built on India's DPI stack — consent-based, no proprietary data, aligned with the DPDP Act 2023
- Product recommender maps the score band to CGTMSE / MUDRA / standard loans — more approvals, safely
- Score history gives IDBI post-disbursement monitoring — origination and monitoring from one platform

List of features offered by the solution

Data Aggregation — 5 sources, one consent

- Udyam registry · GST (GSTR-1 / 2A / 2B / 3B, HSN, credit notes, ITC) · ITR (ITR-4 / 5 / 6) · Account Aggregator bank statements (Finvu) · MCA company, charges & DIN-verified directors
- GST ↔ ITR ↔ bank-credit reconciliation with a cross-source anomaly flag for officer review

AI / ML Scoring Engine

- Six weighted pillars, 0–1000: Revenue Vitality 25% · Cash Flow Health 20% · Transaction Trust 15% · Compliance 15% · Business Stability 15% · Debt Serviceability 10%
- Rules engine + ML.NET LightGBM calibration (0.6 / 0.4), SSA cash-flow trend, PFI explanations, neutral baseline when a source is missing

Financial Health Card & Report

- Score gauge, 6-pillar radar, top strengths & risks, lending eligibility — working-capital limit (Nayak) + term-loan capacity, 14-ratio appraisal
- Product recommendation (CGTMSE / MUDRA / standard), Score History trend, printable Financial Report, email delivery

Bank Portal — Customer 360 (new for the Finals)

- Portfolio list with scores & bands, bank dashboard, per-MSME tabs: Score, GST, Bank (AA deep analysis), ITR, MCA, Directors

Hand-off to the lender (new for the Finals)

- Raise a scored MSME as a loan case — IDBI LOS (appraisal case) · ULI (OCEN 4.0 derived data packet) · ONDC (beckn FIS12); the officer previews the exact payload before confirming
- Append-only push audit with score, band and eligibility frozen at the moment the case was raised; pointing a channel at a live endpoint is configuration, not code

Security & Language

- Email-OTP login, CSP security headers, source-IP fraud audit; 7 Indian languages via Bhashini; Groq AI assistant

What's new since Round 1 — built for the Finals

Score History & Trend

- Every analysis is saved (append-only), never overwritten
- Trend chart on the Health Card and in the bank portal
- The base for post-disbursement early warning

Bank Portal · Customer 360

- Portfolio list with scores & bands, bank dashboard
- Per-MSME tabs: Score, GST, Bank, ITR, MCA
- Director drill-down with DIN verification

Loan case → LOS · ULI · ONDC

- Raise a scored MSME as a case from Customer 360
- Preview the exact payload, then confirm — never one click
- A live endpoint is configuration, not a code change

MCA + DIN — 5th data source

- Company status, capital, charges & AGM compliance
- Directors verified against DIN with individual PAN
- Follows entity type — shown only for companies

ITR now inside the score

- Filing timeliness, e-verification, tax paid ratio
- 44AB audit and tax-demand signals
- GST ↔ ITR turnover reconciliation

Lending, Report & Access

- Working-capital limit (Nayak) + term-loan capacity
- 14-ratio appraisal, printable report, email delivery
- 5 seeded demo MSMEs · 7 languages · IDBI palette

Process flow diagram or Use-case diagram

1 · Onboard & log in

1. Enter Udyam number

Udyam registry lookup fetches the enterprise profile

2. Review & consent

Owner confirms details, email + mobile, gives consent and registers (mobile seeded in the Finvu SimBanks sandbox)

3. Email OTP → dashboard

OTP verified, session starts, routed by status

Existing MSME

/Auth/CustLogin → email → OTP → dashboard

2 · Collect data (one button each)

4a. GST

12 months × 7 return files — GSTR-1, 2A, 2B, 3B, HSN, credit notes

4b. Bank statements (AA)

Finvu consent → owner approves → FI fetch of the statement

4c. ITR

Filing, income, tax paid, audit

4d. MCA + DIN

Companies only — status, charges, directors

Stored per MSME

each source saved as JSON in the <UAN>/blob folder; fetched sources are locked

3 · Score

5. Analyze

Five extractors read every file; a missing source redistributes its weight (neutral baseline)

6. Rules 0.6 + LightGBM 0.4

Six dimensions · 0–1000 · band · PFI reasons · GST-ITR-bank anomaly · lending eligibility & products

7. Persist

result.json → Blob · score summary → SQL · score history appended → SQL

Re-analyze

any time — re-runs 5–7 from the stored files; every run adds a history point

4 · Use the score

8. Financial Health Card

Gauge, 6-pillar radar, strengths & risks, products, Score History

9. Financial Report

Printable appraisal — print or email

10. Bank Portal

Bank login → customer list → Customer 360: Score, GST, AA, ITR, MCA, Directors

11. Raise the loan case

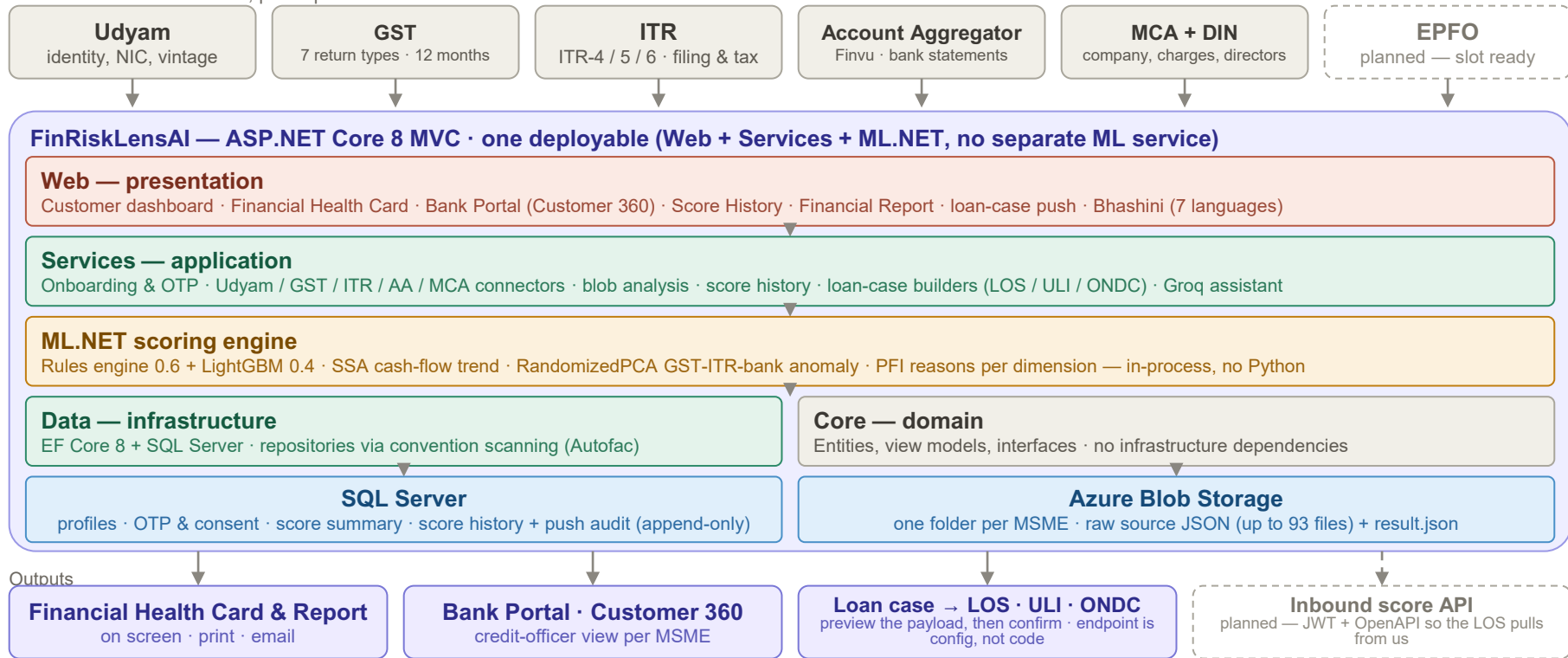
Push to LOS · ULI (OCEN 4.0) · ONDC (beckn) — preview the payload, then confirm

Inbound API — planned

JWT + OpenAPI so the LOS pulls a score from us; the outbound push above is built

Architecture diagram of the proposed solution

Data sources — one consent, pulled per MSME into its own blob folder



Technologies to be used in the solution

Backend — Orchestration & API

- ASP.NET Core 8 MVC (C#) — connectors for Udyam, GST, ITR, Account Aggregator (Finvu), MCA / DIN; outbound loan-case push to LOS / ULI / ONDC
- ML.NET in-process — scoring, feature extraction and explanations in the same deployable; no separate ML service
- Autofac DI with convention scanning · Serilog structured logging · CSP security headers · DataProtection keys persisted

AI / ML Stack

- ML.NET LightGBM calibration — final score = 0.6 rules engine + 0.4 ML model, trained on 3,000 MSME profiles
- ML.NET SSA time-series — 12-month cash-flow trend · RandomizedPCA — GST vs ITR vs bank anomaly detection
- Permutation Feature Importance — per-dimension, audit-ready reasons · Groq LLM — AI assistant for questions on the score

Data & Storage

- SQL Server + EF Core 8 — MSME profiles, OTP & consent, score summary, append-only score history and loan-case push audit; SQL-backed sessions
- Azure Blob Storage — one folder per MSME: raw source JSON (up to 93 files) + result.json

Infrastructure & DevOps

- Single deployable web app (modern monolith) — IIS / Azure App Service / AWS; GitHub source control
- Planned: inbound JWT + OpenAPI spec for LOS score pull; Azure API Management as the ULI / OCEN gateway

Technologies to be used in the solution

Frontend

- ASP.NET Core MVC + Razor views, Bootstrap 5 — server-rendered, responsive; two themes including IDBI's official palette
- Chart.js — score gauge, 6-pillar radar, score-history trend, cash-flow charts · ApexCharts — GST analysis; charts follow the theme switch
- Bhashini website-translation plugin — English + Hindi, Marathi, Tamil, Gujarati, Konkani, Sanskrit

DPI & Data Integrations

- Account Aggregator (Finvu sandbox, SimBanks FIP) · GST System API · Udyam Registry · ITR · MCA 21 (company, charges, DIN)
- Outbound loan case built — IDBI LOS · ULI (OCEN 4.0) · ONDC (beckn FIS12) · inbound DSP score API and EPFO — planned

Security

- Email-OTP customer login and separate bank-admin login, SQL session cache, CSP headers, source-IP fraud / VPN / Tor audit with geolocation

Estimated implementation cost (optional)

→ **22 L**

One-time setup (build + security)

→ **31 L / yr.**

Recurring (cloud + data + support)

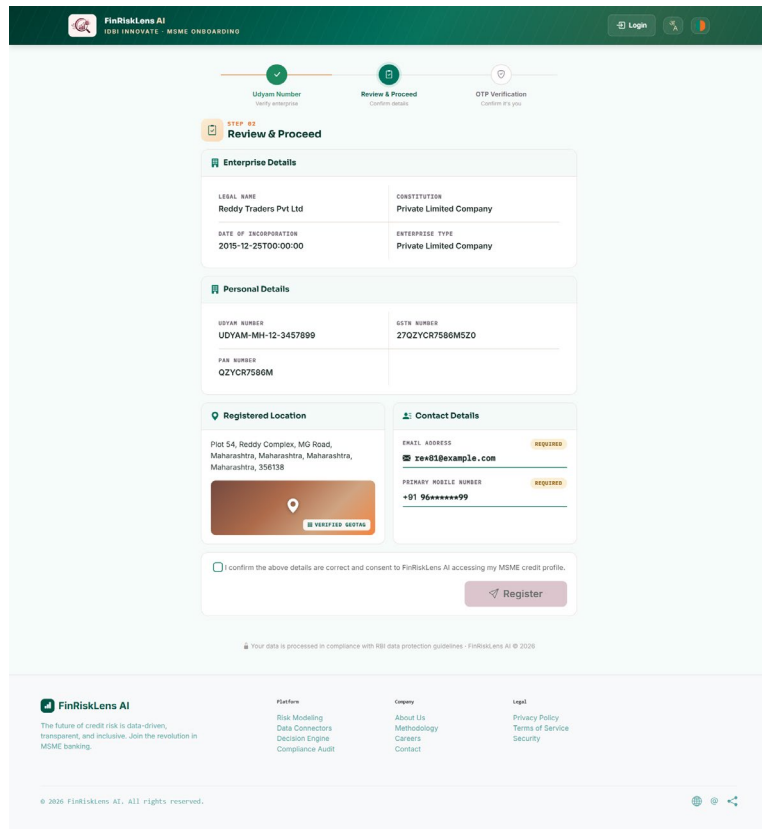
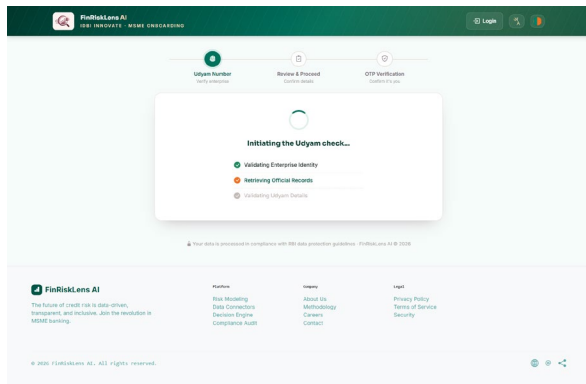
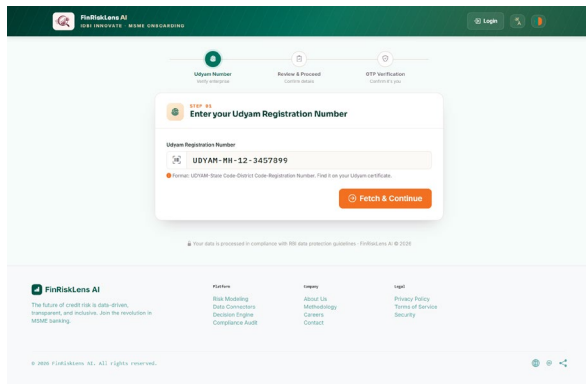
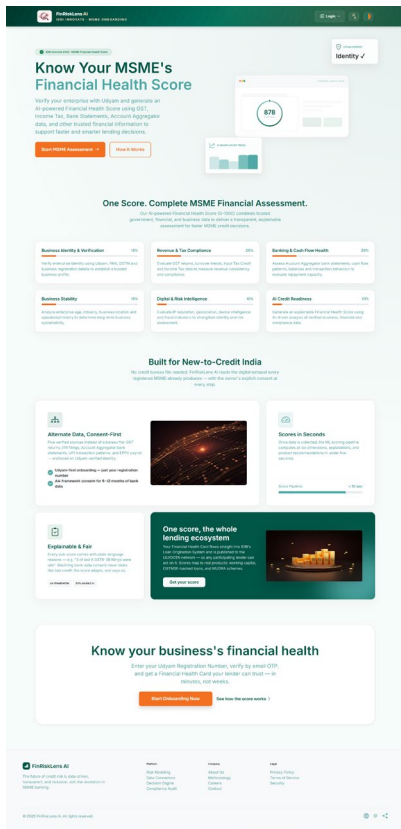
~ → **1.15 Cr**

3-year total cost of ownership

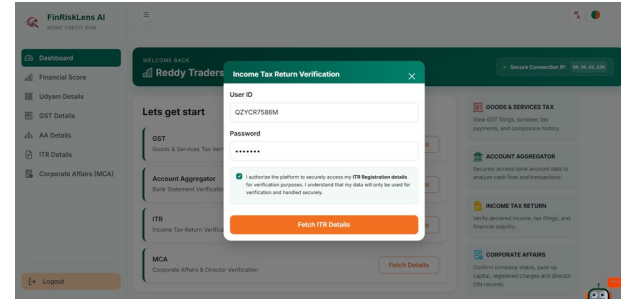
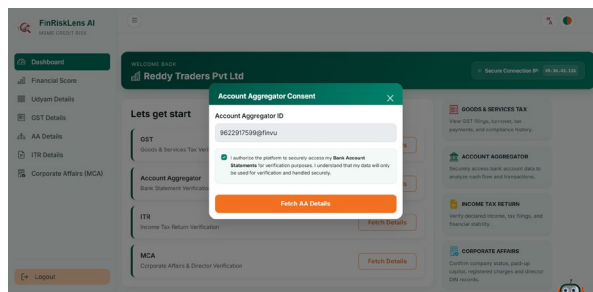
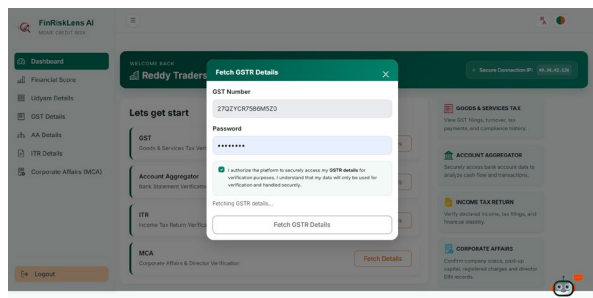
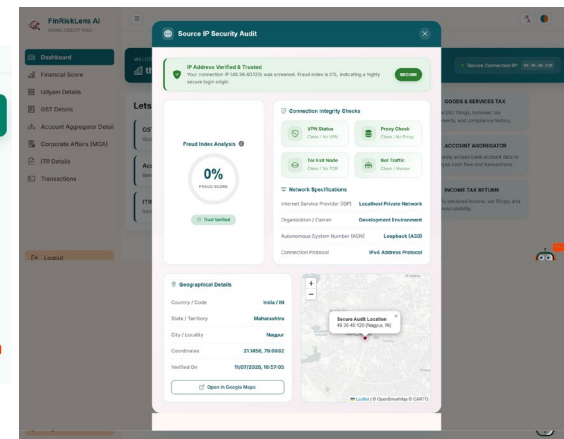
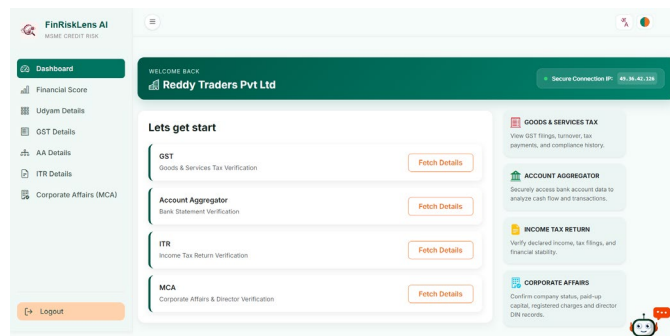
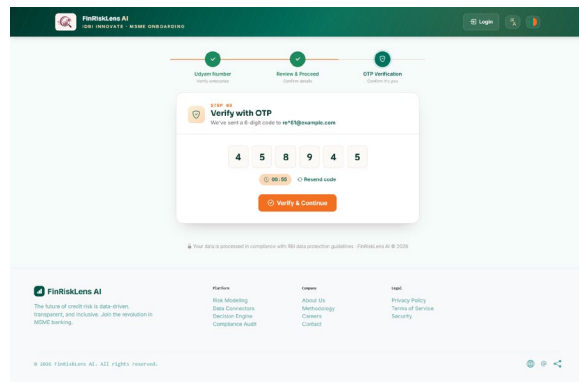
Cost breakdown - centralized cloud deployment serving all IDBI branches pan-India

Cost head	Type	Estimate (INR)
Development - 3-member team, ~6-month build	One-time	→18 L
Security testing & RBI / DPDP compliance audit	One-time	→4 L
Cloud & servers - Azure App Service, SQL, Blob, APIM	Annual	→12 L
Data & API usage - AA (Finvu), GST, ITR	Annual	→10 L
Maintenance, support & ML model retraining	Annual	→9 L

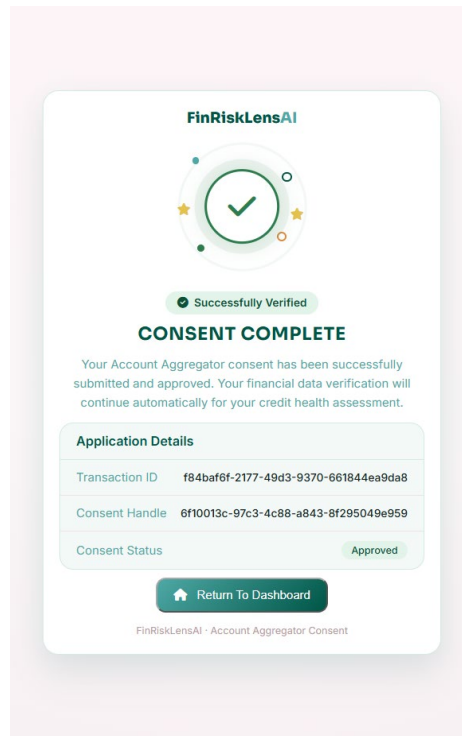
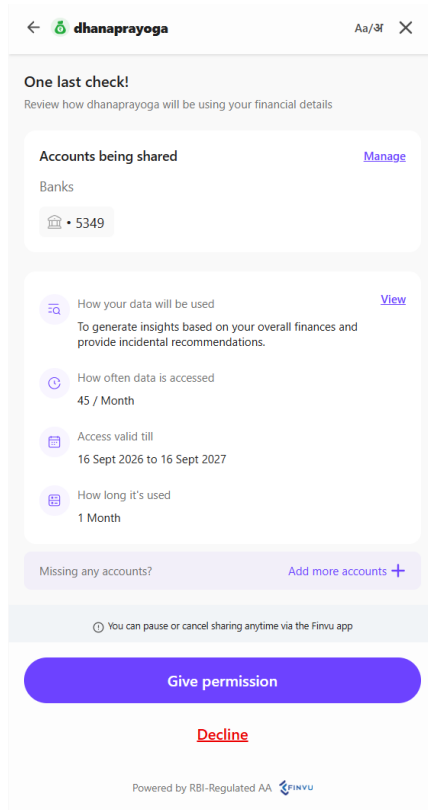
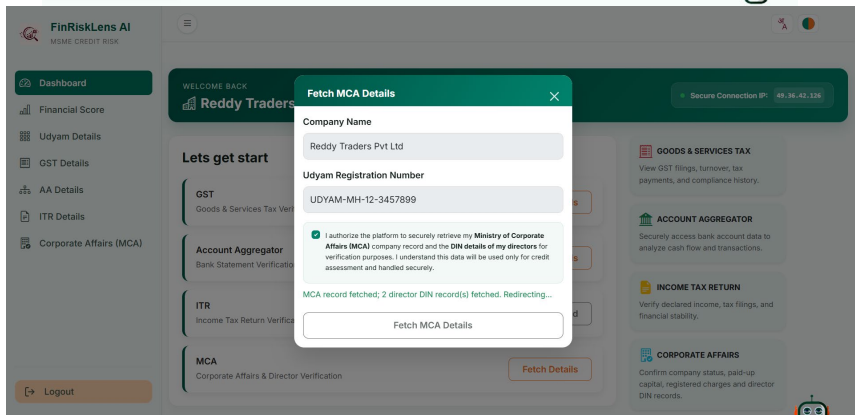
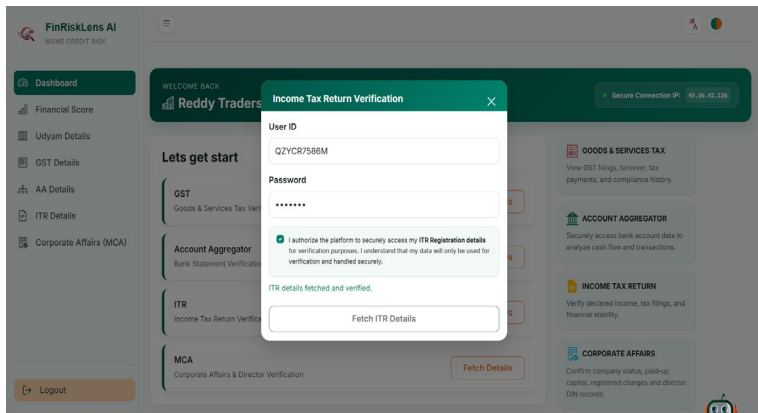
Snapshots of the prototype



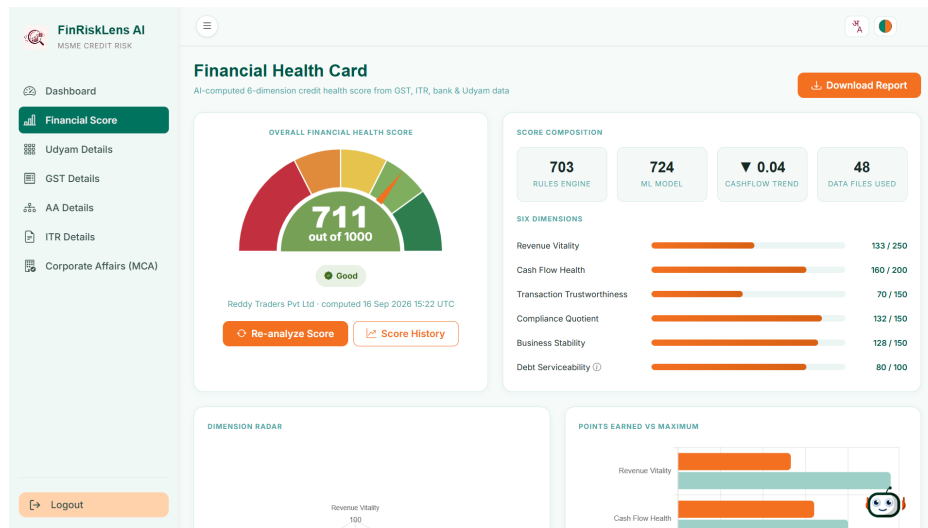
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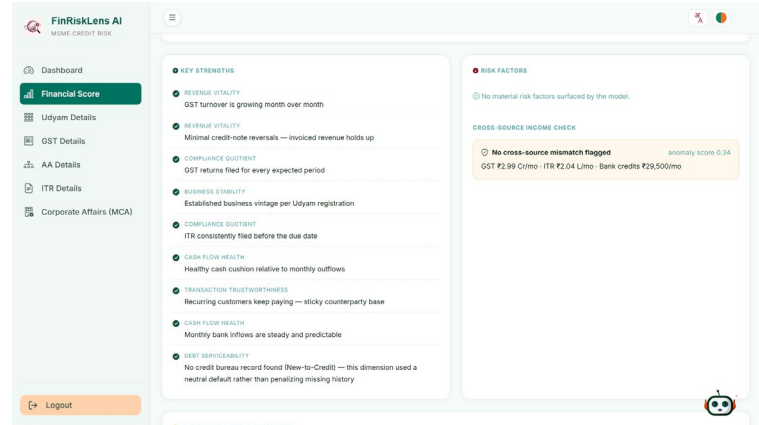
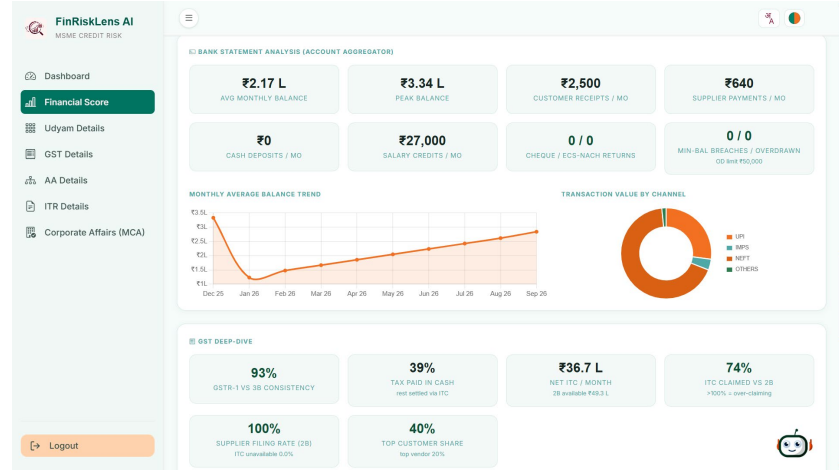
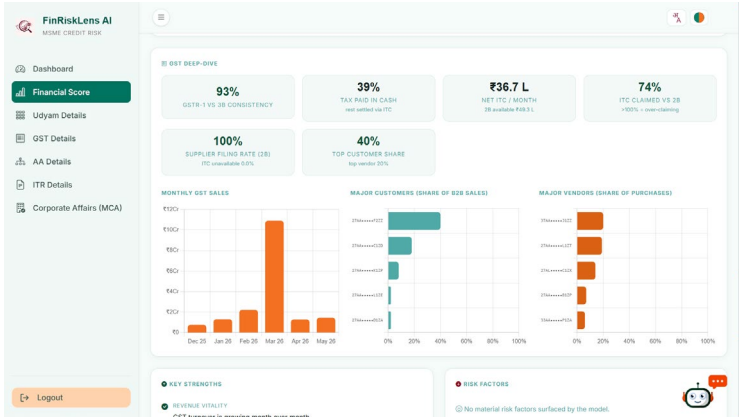
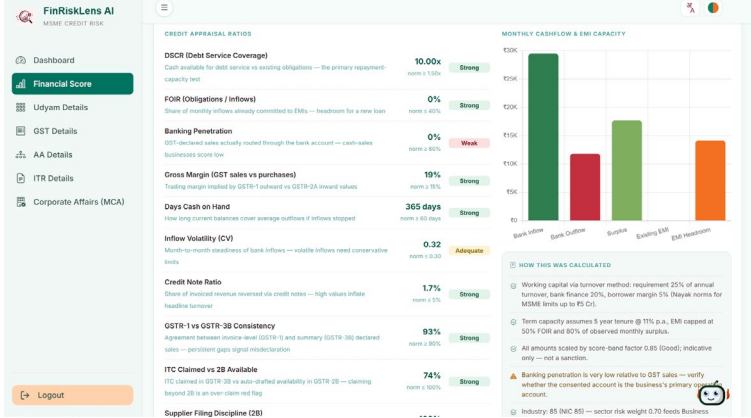
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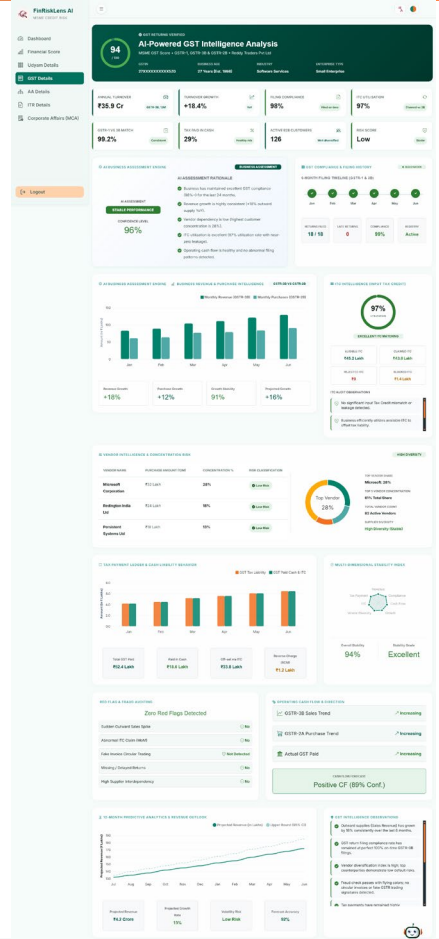
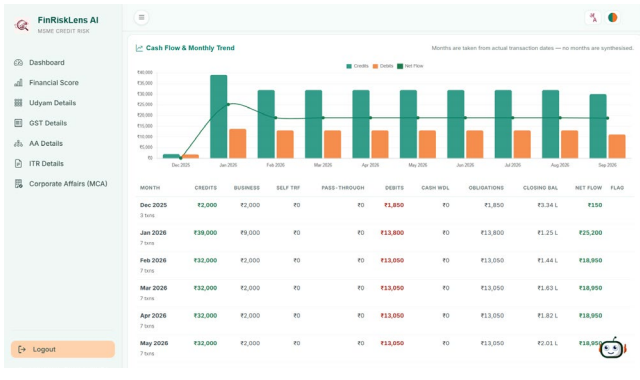
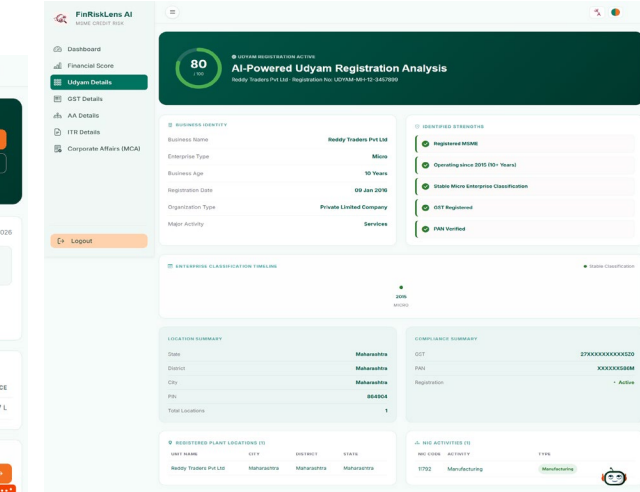
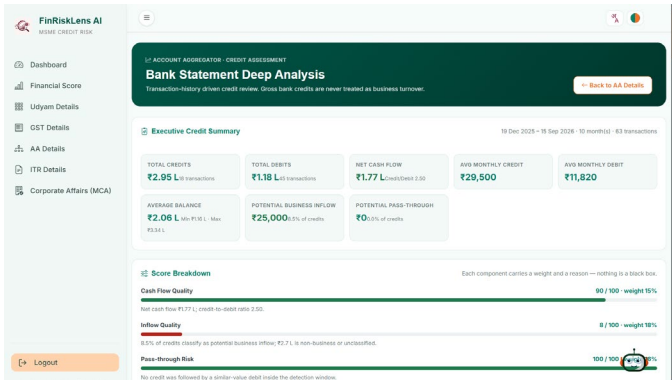
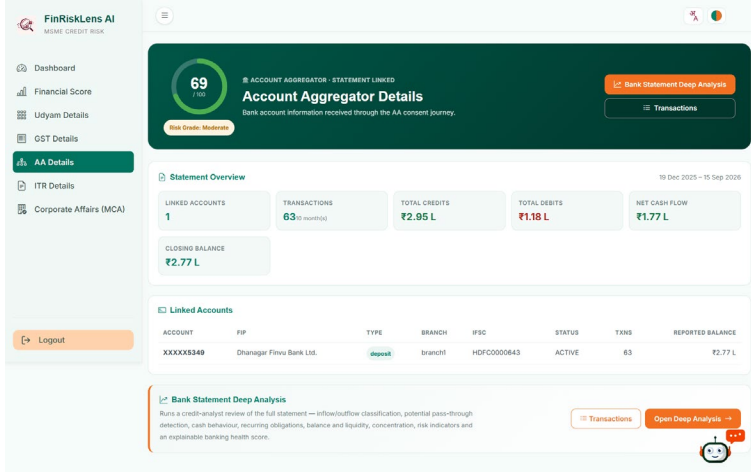
Snapshots of the prototype



Snapshots of the prototype



Snapshots of the prototype



Snapshots of the prototype

FinRiskLens AI
AI-DRIVEN COMPLIANCE RISK

Dashboard
Financial Score
Udyam Details
GST Details
AA Details
ITR Details
Corporate Affairs (MCA)

Income Tax Return Analysis
ITR-5 - PROCESSED
Ruddy Traders Pvt Ltd PAN: XXXXXX588M AY 2026-27
95% (90)
₹24.47 L GROSS INCOME ₹7.68 L TAX + INTEREST ₹1.16 Cr ON TIME FILING LOW RISK

Overview Income & Tax Financials Compliance

FILING SNAPSHOT
Form: ITR-5
Assessment Year: 2026-27
Financial Year: 2025-26

ASSEESSE PROFILE
Name: Ruddy Traders Pvt Ltd
Constitution: Partnership Firm
PAN: XXXXXX588M
GSTIN: 27XXXXXXX3329
Formed On: 16 Sep 2019
Registered Address: 47, Ruddy Complex, MG Road, Maharashtra, Maharashtra, 404904
Email: ruddy58@example.com
Acknowledgement: 490837518959643

AI OBSERVATIONS
1. ITR-5 for AY 2026-27 was filed on time under section 139(1).
2. Return is e-verified and processed by CPC, so the declared figures are on record with the department.
3. Tax liability is fully discharged with no outstanding demand.
4. Interest of ₹7,876 under sections 234B/234C indicates advance tax was short-paid or paid late — a cash-planning signal rather than evasion.
5. Tax audit completed by F. A. Deshpande & Co. (UDIN 26926438AAAAA3109), giving the reported financials third-party assurance.
6. GST and ITR turnover agree within 2.88% — the two filings tell the same revenue story.
7. Net margin of 18.09% on turnover of ₹1.16 Cr is healthy for an MSME.

STATUS
Filed On: 03 Oct 2026
Due Date: 31 Oct 2026
Return: Processed

INCOME SUMMARY
Business / Profession: ₹2,260,546
House Property: ₹0
Capital Gains: ₹0
Other Sources: ₹180,000
Gross Total Income: ₹2,440,546
Chapter VI-A Deductions: ₹12,000
Total Income: ₹2,434,546

BANK ACCOUNTS ON RETURN
Bank: Bank of Maharashtra
Account: XXXXXX0026
IFSC: VU0R037817
Type: Current
Refund: Yes

Logout

FinRiskLens AI
AI-DRIVEN COMPLIANCE RISK

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Overview Income & Tax Financials Compliance

PROFIT & LOSS — 01-04-25 TO 31-03-26 (Q1)
₹1.16 Cr REVENUE FROM OPERATIONS
₹1.8 L OTHER INCOME
₹99.22 L TOTAL EXPENSES
₹28.58 L PROFIT BEFORE TAX
₹20.98 L PROFIT AFTER TAX
18.09% NET MARGIN

BALANCE SHEET — 31 MAR 2026
Partner / Member Capital: ₹2,600,000
Total Liabilities: ₹4,920,000
Total Assets: ₹6,310,000

KEY RATIOS
Current from the figures on this return only — a dash means the return does not carry the source file.
Profit Before Tax Margin: 24.64%
Net Profit Margin: 18.09%
Asset Turnover: 1.94x
Debt to Equity: 0.89x

PARTNERS
Anil Deshpande (90% Ownership)
Heena Bhat (10% Ownership)

Profit Share: 89%
Capital Balance: ₹90 L
Working Partner: Yes

Profit Share: 37%
Capital Balance: ₹10 L
Working Partner: Yes

Remuneration to Partners: ₹448,293
Interest on Capital: ₹162,880
Allowable (u/ 40B): ₹591,179
Disallowed: ₹0

Logout

FinRiskLens AI
AI-DRIVEN COMPLIANCE RISK

Dashboard
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Udyam Details
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₹24.47 L GROSS INCOME ₹7.68 L TAX + INTEREST ₹1.16 Cr ON TIME FILING LOW RISK

Overview Income & Tax Financials Compliance

FILING & TAX HEALTH RECAP
Each component carries a weight and a reason — nothing here is a black box.
Filing Timeliness: 98/100
E-Verification: 96/100
Processing Status: 96/100
Tax Discharged: 95/100
Audit Compliance: 96/100

VERIFICATION MATRIX
1. PAN present on return: GST/linked
2. Filed by due date: E-verified
3. Processed by CPC: No outstanding demand
4. Tax fully discharged: Audit obligation met
5. GST turnover reconciles: Refund bank account on record

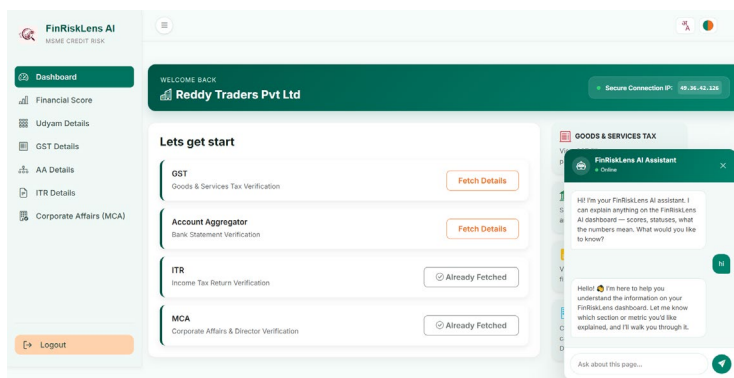
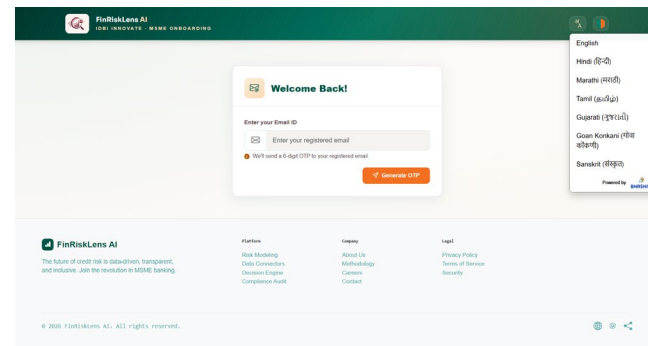
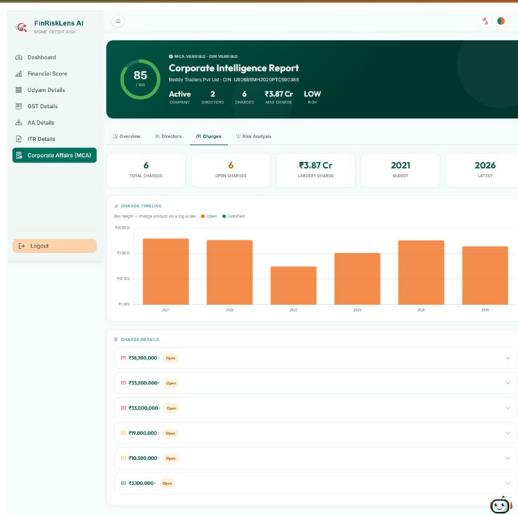
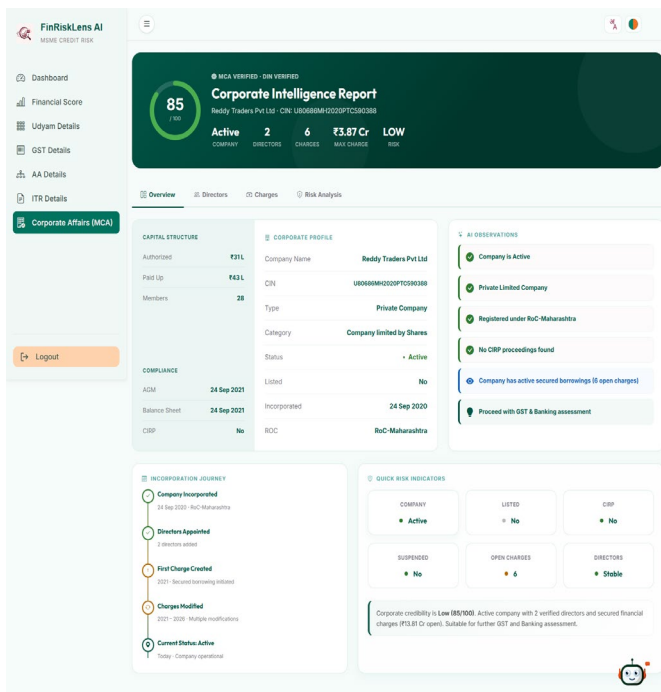
FILING & VERIFICATION
Filing Type: Original
Filing Section: 139(1)
Filing Mode: Digital Signature Certificate (DSC)
E-Verification: Verified (03 Oct 2026)
Verified By: Anil Deshpande, Managing Partner
Verified On: 03 Oct 2026
Place: Maharashtra

TAX AUDIT
Section: 44AB
Auditor: F. A. Deshpande & Co.
Membership No.: 068196
UDIN: 26926438AAAAA3109
Firm: SCA/SCB Firm
03 Sep 2026

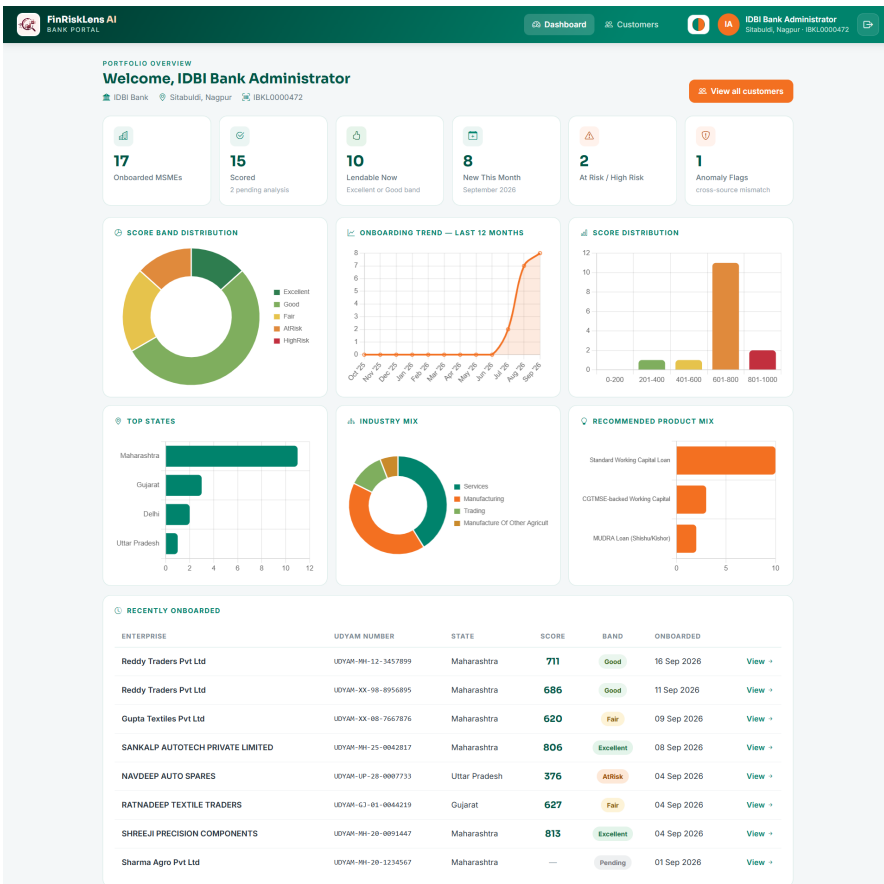
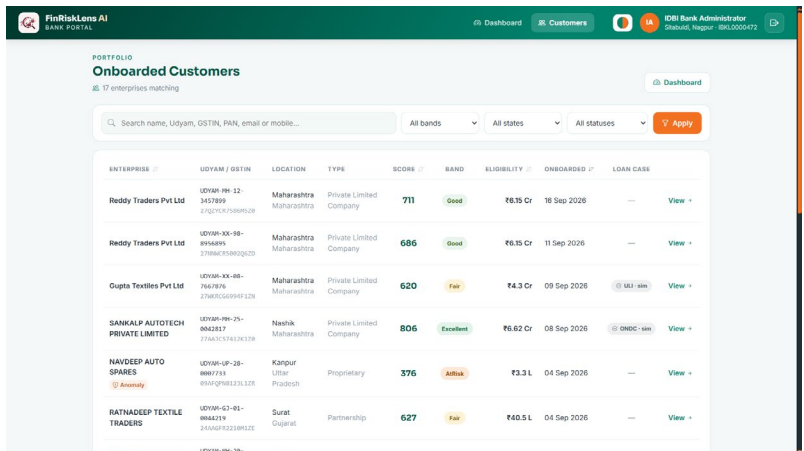
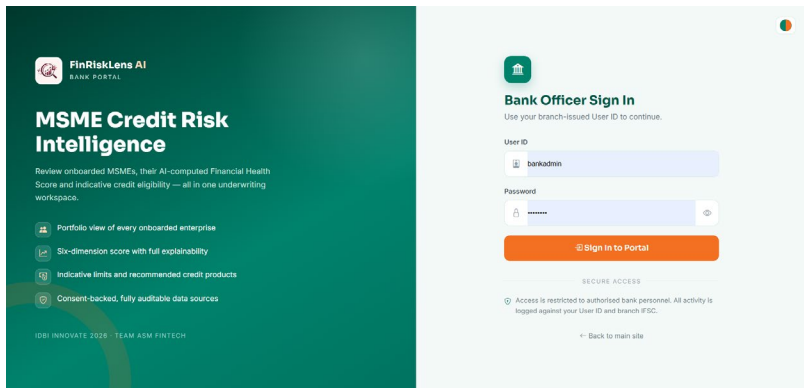
GST — ITR TURNOVER RECONCILIATION
Turnover declared to the GST department against turnover declared on this return. A wide gap is a clear integrity signal worth resolving before filing, not proof of wrongdoing.
₹1.19 Cr GST REPORTED ₹1.16 Cr ITR DECLARED ₹3.34 L VARIANCE 2.88% VARIANCE

Logout

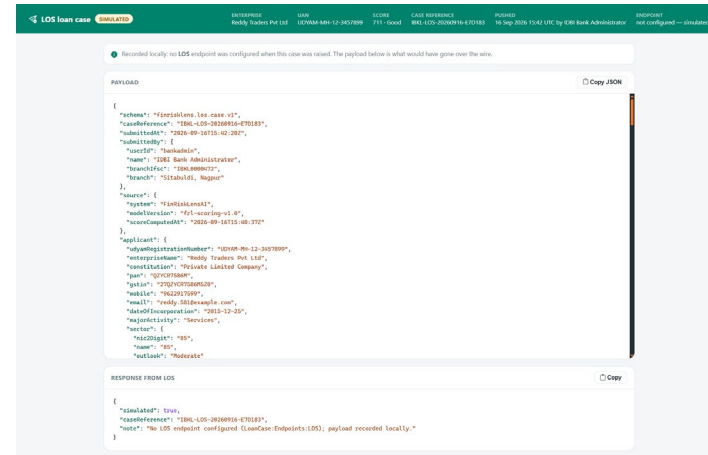
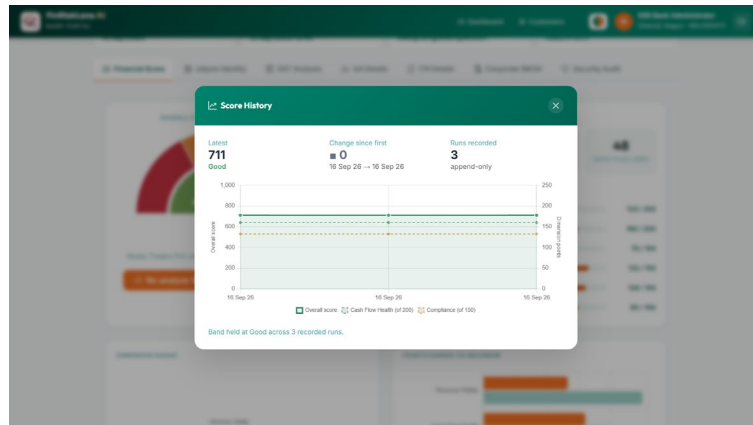
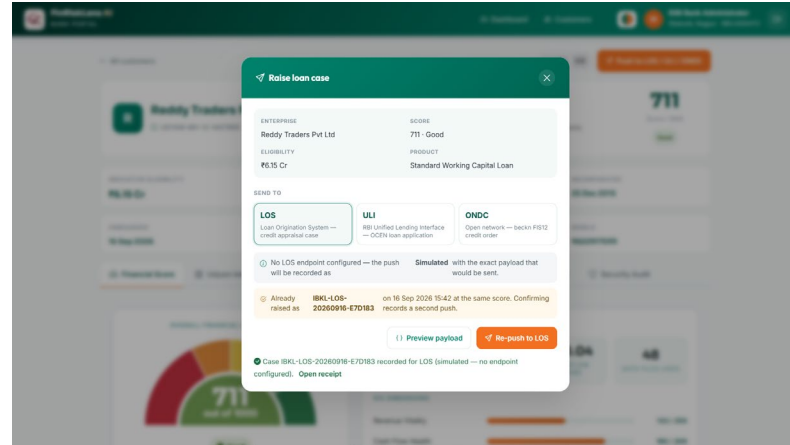
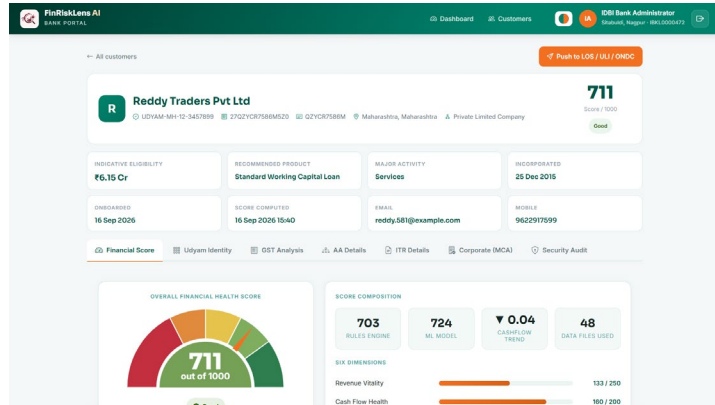
Snapshots of the prototype



Snapshots of the prototype



Snapshots of the prototype



Prototype Performance report/Benchmarking

715/1000

Live score on a real MSME — 12 months of GST + bank data (band: Good)

< 10 s

End-to-end scoring (sub-second when warm)

5

Data sources fused: Udyam · GST · ITR · AA · MCA (+ EPFO ready)

3000

MSME profiles the LightGBM model trains on

Benchmark vs. traditional credit assessment

Metric	Traditional / bureau	FinRiskLensAI
Turnaround time	2-3 weeks (manual)	Seconds (automated)
NTC / NTB borrowers	Rejected - no history	Scored on alternate data
Data sources	1 - bureau file	5 - Udyam, GST, ITR, AA, MCA (+ EPFO ready)
Explainability	Black-box score	PFI reason per dimension
Fraud / integrity check	None	GST-ITR-bank anomaly + MCA / DIN checks
Monitoring	One-time assessment	Score history + trend on every analysis
Hand-off to the lender	Manual file / PDF	LOS, ULI/OCEN or ONDC payload in one action

Additional Details/Future Development (if any)

Next Data Integrations

- EPFO — employee count & PF-remittance regularity (the feature slot is already built)
- GST e-invoice / e-way bill signals and CERSAI collateral lookup
- ONDC / GeM / UPI merchant-settlement data for micro-enterprises

Monitoring, Fraud & Compliance

- Early Warning System on score history — band drop, GST filing lapse, new MCA charge, bounce spike
- Sanctions / PEP screening, RBI fraud registry & wilful-defaulter match
- DPDP consent ledger across all sources with customer-side revoke & auto-purge

Lending Ecosystem & Integration

- Inbound API — JWT + OpenAPI so your LOS pulls a score from us (the outbound case push is already built)
- ULI DSP + OCEN certification and a live ONDC subscriber ID; CIBIL / CRIF bureau blend for NTB borrowers
- GST-invoice financing and co-lending workflows with NBFC partners

Decision & Governance

- Credit-policy console — Approve / Refer / Decline with editable product-wise cut-offs & knock-outs
- PDF Credit Appraisal Memo with data-provenance appendix (source, time, consent ID)
- Model card & governance page (RBI model-risk); portfolio analytics for IDBI teams

Provide links to your:

- GitHub Public Repository : <https://github.com/abhinavmukwane/FinRiskLensAI>
- Demo Video Link (3 Minutes): <https://video.finrisklensai.com> OR [Google Drive Link](#)
- Final Product Link: <https://finrisklensai.com>

Demo accounts — customer login at /Auth/CustLogin (enter email → OTP is shown on screen) · bank portal at /Auth/BankLogin,

shreeji.precision@finrisklens.demo : UDYAM-MH-20-0091447 · 813 Excellent · Shreeji Precision Components

sankalp.autotech@finrisklens.demo : UDYAM-MH-25-0042817 · 806 Excellent · Sankalp Autotech Pvt Ltd (full MCA + DIN)

ratnadeep.textiles@finrisklens.demo : UDYAM-GJ-01-0044219 · 627 Fair · Ratnadeep Textile Traders

navdeep.autospares@finrisklens.demo : UDYAM-UP-28-0007733 · 376 At Risk · Navdeep Auto Spares

This together cover the whole score range — approve, judge, and decline — with every data tab populated.

Bank Login: User ID : **bankadmin** | Password : **Bank@123**



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Thank you!